NISM SERIES XXI-B · SEPTEMBER 2026 WORKBOOK · STUDY NOTES

Portfolio Managers

Run other people's crores like a fiduciary: rules, risk, returns and the maths that proves you earned your fee.

A Portfolio Management Service (PMS) is like hiring a personal chef for your money instead of eating at a buffet (a mutual fund). The PMS firm builds and runs a portfolio in your own name, for a minimum ticket of ₹50 lakh, and can be discretionary (chef decides the menu), non-discretionary (you decide, chef cooks) or advisory (chef only suggests). Because these managers hold huge sums in a fiduciary capacity, SEBI wants the people at the top, the Principal Officer and anyone with fund-management decision power, to prove they know investments, risk, regulation and performance measurement. This exam is that proof.

📋 Exam pattern

150	180 min	60%	25% of the marks assigned to the question
Total marks	Duration	Pass mark	Negative marking

90 MCQs + 6 caselets (30 sub-questions) · Pass mark is 90 out of 150. Test-centre workstations have Microsoft Excel or LibreOffice Calc for numericals, so practise IRR, YIELD, PRICE and DURATION functions. Caselet sub-questions carry 2 marks each and lose 0.5 mark when wrong.

👥 Who needs it

- Principal Officer of a SEBI-registered portfolio manager (must hold the relevant NISM certification specified by SEBI, plus qualification and 5 years' experience)
- Any PMS employee with decision-making authority related to fund management (same qualification, experience and certification as the Principal Officer)
- PMS distributors, who must hold valid certification as specified by SEBI at all times (Code of Conduct for PMS Distributors) and register with APMI
- Other employees of PMS firms who want the common minimum knowledge benchmark the exam sets

🧭 Exam strategy

- Budget time: about 80 minutes for the 90 MCQs and 90 minutes for the 30 caselet sub-questions, plus 10 minutes to review.
- Caselet sub-questions are worth 2 marks each (60 marks in total). Read the case once, note all numbers, then solve the sub-questions in order because they often build on each other.
- Keep Excel open: use IRR/XIRR for MWRR, YIELD/PRICE/DURATION for bonds, and SQRT/NORM.S.INV checks for VaR.
- Negative marking is 25%: guess only when you can eliminate at least two options (a wrong caselet answer costs 0.5 mark).
- Chapters 7, 8, 11, 17, 18, 19 and 20 carry 80 of 150 marks. Master them first.
- Regulatory questions turn on exact numbers and words ('shall not', 'except advisory'). Learn the number table cold.
- For fee caselets, follow the workbook order: gross value, minus expenses and fixed fee, compare with hurdle and high water mark, then apply performance fee, then exit load.
- When asked Sharpe vs Treynor, check how diversified the investor's total wealth is.
- Watch units: annual vs daily σ (√250), percent vs decimal, and semi-annual YTM in modified duration.



Chapter 1 · Weight: 3 marks

Investments

Why people invest, how investing differs from saving and speculation, and how the required return is built: real risk-free rate, then inflation, then a risk premium. It also maps the investment universe and the channels (direct, advisers, PMS, funds) for getting there.

> **MAKE IT STICK**
> Lending a friend ₹1,000 and asking for ₹1,020 back is the pure rate of interest at work. Ask for more because prices will rise, and more again because your friend is unreliable, and you've just built the required rate of return: real rate, then inflation, then risk premium.

1. Saving = income minus spending; investment = committing savings for a period expecting a higher amount back
2. Speculation: taking risk not commensurate with return, with minimal research on true value; it is not defined only by short holding period
3. Investment objectives: capital preservation, capital appreciation, current income, tax saving; objectives must be stated in terms of risk AND return
4. Required return = real risk-free rate, plus compensation for expected inflation, plus a risk premium
5. Nominal rate = (1 + real rate) x (1 + expected inflation) - 1, e.g. (1.02 x 1.06) - 1 = 8.12%
6. Real risk-free rate depends on the desire for current consumption (subjective) and investment opportunities / real growth of the economy (objective)

7 Risk is 'known uncertainty': dispersion around expected return; uncertainty is when causal factors are unknown

8 Business risk comes from sales volatility and operating leverage; financial risk comes from debt (financial leverage)

9 Liquidity risk: difficulty converting to cash near economic worth (T-bills ≈ none; art = high)

10 Other risks: exchange rate, political (country), geopolitical, regulatory

11 Risk and return are positively related, though not linearly

12 Financial investments can be in security form (tradable: shares, bonds) or non-security form (FDs, insurance: no secondary market)

Nominal rate formula
(1 + real) x (1 + inflation) – 1

Workbook example
2% real + 6% inflation = 8.12% nominal

Required return components
Real RFR + inflation premium + risk premium

🎯 EXAM TIP
Expect a nominal/real rate calculation and 'which risk is this?' questions (business vs financial vs liquidity).

⚠️ TRAP
Do not simply add real rate and inflation (2% + 6% = 8%). The workbook multiplies: 8.12%.



Chapter 2 · Weight: 2 marks

Introduction to Securities Markets

How the primary market raises fresh capital (IPO, FPO, rights, private placement, QIP, OFS) and how the secondary market gives liquidity, with clearing corporations guaranteeing exchange trades. It also lists the intermediaries and participants.

MAKE IT STICK
The primary market is the car showroom where new cars (shares) are sold by the maker. The secondary market is the used-car market where owners sell to each other. The clearing corporation is the escrow agent who makes sure every buyer gets the car and every seller gets paid.

1 Primary market: issuers raise fresh capital; secondary market: investors trade among themselves, providing liquidity

2 Private placement under the Companies Act 2013: offer to less than 200 persons

3 QIP: private placement by a listed company to Qualified Institutional Buyers (QIBs)

4 OFS: existing shareholders sell; no increase in share capital; proceeds go to the sellers, not the company

5. Bonus issue: no consideration; made from free reserves; cannot be made if the company has defaulted on debt or FD payments; cannot be withdrawn once announced

6. Anchor investor: a QIB applying for ₹10 crore or more in a book-built public issue (introduced 2009)

7. ADRs list in the USA, GDRs list outside the USA, IDRs are issued in India on foreign shares

8. Clearing corporation provides full novation: buyer to every seller and seller to every buyer

9. Two depositories: NSDL and CDSL; Depository Participants are their agents

10. Custodians hold funds and securities, mainly for institutions and FPIs, and settle trades

Private placement	Anchor investor minimum	Depositories
Offer to < 200 persons	**₹10 crore application**	**NSDL and CDSL**

🎯 EXAM TIP
Definitions of OFS, QIP, rights, bonus and the clearing corporation's novation role are favourite one-liners.

⚠️ TRAP
An OFS is a primary-market route but a secondary-market transaction: the company gets no money.



Chapter 3 · Weight: 5 marks

Investing in Stocks

Equity as an asset class, its risks, and how analysts value shares: DCF (FCFF/FCFE), dividend discount, and relative multiples like P/E, PEG, P/S and EV/EBITDA. It also touches technical analysis and corporate governance.

MAKE IT STICK
Valuing a stock is like valuing a mango orchard: you can count the future mangoes and discount them (DCF), or check what similar orchards sold for per tree (multiples). A PEG below 1 is like an orchard priced cheaply for how fast it is growing.

1. Equity risks: market, sector-specific, company-specific, liquidity, currency

2. Three valuation approaches: discounted cash flow, asset-based and relative (multiple-based)

3. FCFF is discounted at WACC to value the firm; FCFE is discounted at cost of equity to value equity

4. Cost of equity (CAPM): Ke = Rf + β(Rm - Rf), where Rf is usually the 10-year government bond yield

5. WACC = Ke x We + Kd x (1 - tax) x Wd

6. Sustainable growth proxy = retention ratio x ROE = (1 - payout) x ROE

7. Dividend discount model: P = D1 / (k - g), with D1 = D0 x (1 + g)

8. Forward P/E = payout / (k - g): rises with growth, falls with required return

9 PEG = P/E ÷ EPS growth; PEG < 1 suggests undervalued, > 1 overvalued

10 EV = market cap of equity + market value of debt - excess cash

11 EVA = NOPAT - (invested capital x cost of capital); MVA = market value minus capital contributed

12 Technical analysis assumes all information is in price and volume; moving averages are the most popular indicator

Gordon model	PEG thumb rule	EV
P = D1 / (k – g)	**< 1 undervalued, > 1 overvalued**	**Mcap + debt – excess cash**

Growth proxy
Retention ratio x ROE

🎯 EXAM TIP

Short numericals: DDM price, PEG verdict, EV, WACC. Know which cash flow goes with which discount rate.

⚠️ TRAP

Use D1 (next year's dividend), not D0, in the dividend discount model. Discounting FCFE at WACC is wrong.

Chapter 4 · Weight: 5 marks

Investing in Fixed Income Securities

Bond features, credit ratings, yield measures and the maths of bond price volatility: duration, modified duration and convexity. Expect Excel-style YIELD/PRICE/DURATION caselets.

MAKE IT STICK

A bond is a see-saw: rates go up, prices go down. A long-maturity, low-coupon bond sits far from the pivot, so it swings the most. A zero-coupon bond sits at the very end of the plank.

1 Government securities carry practically no default risk, hence 'gilt-edged'

2 Current yield = annual coupon ÷ current market price

3 YTM assumes all coupons are reinvested at the YTM; reinvestment risk is the risk this fails

4 Bond prices and interest rates move inversely

5 Lower coupon means higher price volatility; zero-coupon bonds show the largest % change

6 Longer maturity means higher volatility, but it increases at a diminishing rate

7 Price changes are asymmetric: a fall in yield raises price more than an equal rise lowers it (convexity)

8 Macaulay duration (years) = weighted average time to recover the price in PV terms; it is ≤ maturity and equals maturity only for zero-coupon bonds

9 Modified duration = Macaulay duration ÷ (1 + YTM per period)

10 % price change ≈ - modified duration x change in yield (decimal)

11 Call option lets the issuer retire bonds early; put option lets holders sell back to the issuer

12 Excel: YIELD, PRICE and DURATION functions; Indian day count is 30/360 (basis 0)

Modified duration
MacD ÷ (1 + YTM/frequency)

Price change
-MD x Δy

Workbook example
MacD 5, YTM 12% semi-annual: MD = 5/1.06 = 4.72

Zero-coupon duration
= its maturity

🎯 EXAM TIP
Numericals on current yield, modified duration and price change; conceptual questions on volatility rules.

⚠️ TRAP
For semi-annual bonds divide YTM by 2 before computing modified duration.



Chapter 5 · Weight: 5 marks

Derivatives

Forwards, futures, options and swaps, how exchange-traded markets manage counterparty risk, and the specific rules for portfolio managers using derivatives.

MAKE IT STICK
A futures contract is a restaurant booking with a cancellation fee collected upfront (margin). The clearing corporation is the restaurant manager who guarantees your table even if another diner fails to show.

1 Forwards are OTC and customised; futures are standardised and exchange-traded

2 Options give the buyer a right (not an obligation); the buyer pays a premium to the seller

3 Swaps exchange cash flows, e.g. fixed for floating interest

4 Counterparty risk in futures is mitigated by the clearing corporation

5 Derivatives are a zero-sum game (ignoring taxes and transaction costs)

6 SEBI mandates physical settlement for individual stock and commodity derivatives; index derivatives are cash-settled

7 Initial margin = SPAN margin + Extreme Loss Margin; it should cover losses in 99% of cases; option buyers pay premium margin

8 Open interest = outstanding unsettled contracts; it is not trading volume

9 Commodity options in India devolve into commodity futures

10 PMS may use derivatives (including for hedging and rebalancing) via recognised stock exchanges, on terms in the PMS agreement

11 A client's total derivatives exposure must not exceed the portfolio funds placed with the PM; PMs cannot leverage client portfolios for derivatives

Initial margin
SPAN + ELM, covers 99% of cases

PMS derivatives cap
Exposure ≤ client's portfolio funds

Settlement
Stock and commodity derivatives physical; index derivatives cash

🎯 EXAM TIP
Questions on what the PMS agreement must say about derivatives and on futures vs forwards vs options.

⚠️ TRAP
PMS can use derivatives, but they cannot leverage the client's portfolio to do so.

Chapter 6 · Weight: 5 marks

Mutual Fund

The trust-based structure of Indian mutual funds (sponsor, trustees, AMC, custodian, RTA), how NAV and pricing work, factsheets, and how performance must be advertised.

MAKE IT STICK
A mutual fund is a shared tiffin service: everyone chips in, the cook (AMC) makes the meal, the trustees taste-check it on your behalf, and the custodian keeps the pantry locked.

1 Mutual funds are constituted as trusts under the Indian Trusts Act, 1882

2 Sponsor creates the trust; trustees protect unit-holders; AMC manages day-to-day; custodian (appointed by trustees) holds assets

3 The AMC is appointed by the sponsor or trustees with SEBI approval

4 NAV = (market value of assets + income − expenses) ÷ units outstanding, declared daily

5 Open-ended schemes transact at NAV-related prices; close-ended units trade on exchanges at market price

6 Factsheet is voluntary (market practice) but SEBI disclosure rules apply to it

7 Schemes older than 3 years advertise CAGR for 1, 3, 5 years and since inception, plus point-to-point value of ₹10,000

8 Liquid/money market schemes may show simple annualised yields for 7, 15 and 30 days

9 Compliance officer reports directly to the AMC head and reports non-compliance immediately to trustees

Legal form	Custodian appointed by	Point-to-point illustration
Trust (Indian Trusts Act, 1882)	**Trustees**	**₹10,000 investment**

🎯 EXAM TIP
Roles in the MF structure and advertisement rules for performance.

⚠️ TRAP
The AMC manages the money but does NOT hold custody of the assets; the custodian does.



Chapter 7 · Weight: 10 marks

Role of Portfolio Managers

Who a portfolio manager is, the three service types, registration requirements under SEBI (Portfolio Managers) Regulations 2020, and the core duties and don'ts. This is the regulatory backbone of the exam.

MAKE IT STICK
Think of a discretionary PMS as a valet who drives your car without asking, a non-discretionary PMS as a driver who goes only where you say, and advisory as a friend in the passenger seat giving directions while you drive.

1. A portfolio manager must be a body corporate (includes LLPs; excludes co-operative societies) registered with SEBI
2. First PMS regulations came in January 1993, even before the mutual fund regulations
3. Discretionary: manager decides (any degree of discretion); non-discretionary: client decides, manager executes; advisory: manager advises, client decides and executes
4. Net worth requirement: ₹5 crore, maintained at all times (not applicable to co-investment PM)
5. Principal Officer: professional qualification (finance, law, accountancy, business management, NISM 1-year PGP or CFA charter), plus 5 years' experience (at least 2 in portfolio management, investment advisory or fund management), plus NISM certification
6. At least one more employee: graduate with 2 years' experience; anyone with fund-management decision authority needs the Principal Officer's qualifications
7. Minimum investment ₹50 lakh per client (not for accredited investors or co-investment PM)
8. Client funds in a separate account in a Scheduled Commercial Bank; each client's securities segregated
9. Cannot borrow funds/securities for clients, cannot lend securities except as permitted, cannot charge distribution fees on direct-plan MF units, cannot leverage for derivatives
10. Cannot invest in portfolios managed by another PM, or on the advice of any other entity; no bill discounting, badla or lending

11 Pooled trades must be allocated pro rata at the weighted average price of the day, with no open positions

12 Off-market transfers allowed only for own settlement, margin, unlisted dealings, specific client consent, or as SEBI specifies

13 Exit load cap: 3% (year 1), 2% (year 2), 1% (year 3), nil after 3 years; no lock-in allowed

14 Custodian mandatory except for advisory-only PMs (and co-investment PMs); records preserved for 5 accounting years

15 Compliance officer cannot be the Principal Officer (except for co-investment PMs) and must report non-compliance to SEBI independently

Net worth **₹5 crore**	Minimum ticket **₹50 lakh**	Principal Officer experience **5 years (2 years in PM/IA/fund management)**
Other key employee **Graduate + 2 years' experience**	Exit load caps **3% / 2% / 1% / nil after 3 years**	Record preservation **5 accounting years**
Audit certificate to SEBI **Within 6 months of close of accounting period**		

🎯 EXAM TIP
Lists of do's and don'ts, net worth, minimum investment, exit-load slabs and Principal Officer eligibility are near-certain questions.

⚠️ TRAP
Advisory-only PMs need not appoint a custodian, but non-discretionary PMs must. 'No lock-in' still allows exit loads within the caps.

Chapter 8 · Weight: 10 marks

Operational Aspects of Portfolio Managers

Who can invest, what the Disclosure Document must contain, onboarding and KYC, the client agreement, grievance redressal and ODR, reporting to SEBI and FIU, and the PMS fee mechanics (high water mark, hurdle rate). Caselets on performance fee are common.

MAKE IT STICK
A high water mark is like a mountain climber's best altitude: the guide (manager) gets a bonus only for climbing above the highest point already reached, not for re-climbing ground lost in a fall.

1 Eligible investors (min ₹50 lakh): individuals, NRIs (per RBI), HUFs, proprietorships, AOPs, partnership firms, LLPs, trusts, body corporates

2 Disclosure Document must be given with the account opening form before or at the time of signing; the old '2 days in advance' rule was dropped in 2020

3. Simplified Disclosure Document format via SEBI circular of 9 September 2025: static Part I and dynamic Part II (performance for last 3 years using TWRR, audit observations of 3 years)

4. Dealing room: recorded lines only, no mobile phones, restricted internet and access

5. PMs with AUM of ₹1,000 crore or more need an automated order management and allocation system with minimal manual intervention

6. Uniform KYC since 1 January 2012 via KRA; CKYCR upload for individual accounts opened from 1 August 2016

7. Client must separately write or type that they understood the fee structure; fee calculation tool and MITC document mandatory

8. Direct onboarding option must be offered with no charges except statutory charges (not for co-investment PMS)

9. Client may invest cash, securities or both worth at least ₹50 lakh; joint-holding benefits go to the first holder

10. Dispute resolution (Reg. 22A, from 3 July 2023): mediation, conciliation or arbitration via SMART ODR; pre-conciliation and conciliation are free for investors

11. Suspicious transactions must be reported to FIU-IND within 7 working days

12. No upfront fees; brokerage at actuals; operating expenses (excluding brokerage) capped at 0.50% p.a. of average daily AUM

13. Performance fee is charged only on gains above the high water mark and, usually, above a hurdle rate

Operating expense cap	Upfront fee	STR to FIU-IND
0.50% p.a. of average daily AUM (excl. brokerage)	**Not allowed**	**Within 7 working days**

Automated system threshold	CKYCR uploads from	Uniform KYC from
AUM ≥ ₹1,000 crore	**1 August 2016**	**1 January 2012**

🎯 **EXAM TIP**
The performance-fee caselet (fixed fee, brokerage, expenses, hurdle, HWM) is a classic. Practise the workbook layout step by step.

⚠️ **TRAP**
When the pre-fee value is still below the high water mark, the performance fee is NIL even if that year's return is positive.



Chapter 9 · Weight: 10 marks

Portfolio Management Process

The planning, execution and feedback loop: Investment Policy Statement, objectives and constraints, investor profiling (BB&K personalities, life-cycle phases), benchmarking, strategic vs tactical asset allocation and rebalancing.

MAKE IT STICK

The IPS is the GPS route for a road trip: destination (objectives), rules of the road (constraints) and how you'll judge the driver (benchmark). SAA is the planned route; TAA is a quick detour to dodge a traffic jam.

1. Steps: (1) policy statement, (2) study current conditions and forecast, (3) construct the portfolio, (4) measure and evaluate performance
2. IPS purposes: realistic expectations, effective decisions, a framework for evaluating the manager, and protection against inappropriate or unethical decisions
3. Objectives are framed in risk, return and liquidity; constraints are liquidity, time horizon, tax, regulatory and unique preferences
4. Emergency cash: usually 2 to 3 months' spending
5. LRS: resident individuals can remit up to USD 2,50,000 per financial year
6. BB&K personalities: Adventurer (confident, impetuous), Celebrity (anxious, impetuous), Individualist (confident, careful), Guardian (anxious, careful), Straight Arrow (centre)
7. Life-cycle phases: accumulation, consolidation, spending, gifting
8. Investment approaches are tagged to exactly one strategy (equity, debt, hybrid, multi-asset); APMI prescribes at most 3 benchmarks per strategy
9. Changing an IA's strategy or benchmark tag requires offering existing clients a no-exit-load exit; past track record cannot be used afterwards
10. SAA = long-term policy mix ('time in the market'); TAA = short-term deviation ('timing the market')
11. Asset allocation explains about 40% of variation in returns across funds and about 90% of a single fund's return variation over time (Brinson, Hood and Beebower)
12. Rebalancing costs: transaction costs and taxes; illiquid assets are harder to rebalance

First step	**Investment Policy Statement**
Emergency cash	**2 to 3 months' spending**
LRS limit	**USD 2,50,000 per FY**
APMI benchmarks per strategy	**Maximum 3**
Asset allocation explains	**~40% across funds, ~90% within a fund over time**

🎯 **EXAM TIP**

BB&K personality identification, life-cycle phases and SAA vs TAA are favourites; also the benchmark-tagging rules.

⚠️ **TRAP**

Celebrity investors want advisers but are the hardest to deal with; Adventurers rarely use advisers.

Chapter 10 · Weight: 5 marks

Taxation

Residential status (individual, HUF, company), scope of total income, capital gains vs business income, holding-period rules, dividend and interest taxation, buy-back rules, and the Section 9A safe harbour for offshore funds managed from India.

MAKE IT STICK

Your tax passport is stamped by days, not by citizenship: stay 182 days and you're resident. An Indian citizen who escapes tax everywhere but earns over ₹15 lakh in India is pulled back as a 'deemed resident' (NOR).

1. Resident if in India 182 days or more, OR 60 days (and 365 days in the preceding 4 years); 60 becomes 120 days for citizens/PIOs with Indian income above ₹15 lakh
2. Deemed resident: Indian citizen with Indian income above ₹15 lakh and not liable to tax elsewhere; always treated as NOR
3. NOR if non-resident in 9 of 10 preceding years, or in India 729 days or less in 7 preceding years
4. Indian company is always resident; a foreign company is resident if its POEM is in India (CBDT guidelines apply above ₹50 crore turnover)
5. Firm/AOP/HUF is resident unless control and management is wholly outside India
6. A PMS client is taxed as if investing directly in the underlying securities
7. Capital gains indicators: intention of appreciation, low frequency, long holding, shown as investments, own funds
8. Listed securities, equity MF units, UTI units, zero-coupon bonds: long term if held more than 12 months
9. Unlisted shares and other securities: long term if held more than 24 months
10. Specified MF (≤35% in Indian equity, acquired on/after 1 April 2023) and market-linked debentures: always short term
11. Interest deduction against dividend income restricted to 20% of gross dividend; TDS 10% on dividends to residents above ₹10,000
12. Buy-back proceeds from 1 October 2024 are taxed as dividend for the shareholder; cost of acquisition becomes a capital loss
13. Section 9A: fund ≥ 25 members, single member ≤ 10%, top 10 or fewer < 50%, residents ≤ 5% of corpus, ≤ 20% in any entity, monthly average corpus ≥ ₹100 crore; manager's profit share ≤ 20%

Resident (basic)	Deemed resident income	Listed shares LTCG period
182 days, or 60 days + 365 days in 4 years	**> ₹15 lakh Indian income**	**> 12 months**

Unlisted shares LTCG period	9A corpus	9A resident participation	9A minimum members
> 24 months	Monthly average ≥ ₹100 crore	≤ 5% of corpus	25

🎯 EXAM TIP
Residency day-counts, holding periods and the Section 9A numbers are the easiest marks here.

⚠️ TRAP
Specified mutual fund units bought on or after 1 April 2023 are always short-term, regardless of holding period.

Chapter 11 · Weight: 10 marks

Regulatory, Governance and Ethical Aspects of Portfolio Managers

PMLA record-keeping, insider-trading code for intermediaries, PFUTP prohibitions, the 2020 PMS Regulations summary (permitted investments, reports, related-party limits, code of conduct), Eligible Fund Managers, distributor code, research objectivity, soft dollars, investor charter and cyber security.

MAKE IT STICK
The regulations are the referee's rulebook. PMLA watches the money's origin, the insider rules guard secrets behind a Chinese wall, PFUTP blows the whistle on market tricks, and the PMS rules make sure the manager never puts his own goals ahead of the client's.

1. PMLA in force from 1 July 2005: report cash transactions above ₹10 lakh, or integrally connected series exceeding ₹10 lakh in a month
2. PMLA records: 10 years (rules) for prescribed transactions; Section 12 records 5 years from transaction or end of relationship
3. Insider trading code: pre-cleared trades executed within 7 trading days; contra trade barred for at least 6 months; profits go to the IPEF
4. 'Material financial relationship': payments of at least 25% of the designated person's annual income in the past 12 months
5. Discretionary PMs invest only in listed/traded securities, money market instruments, MF units and SEBI-specified securities
6. Non-discretionary and advisory PMs may invest up to 25% of AUM in unlisted securities; up to 100% for large value accredited investors
7. MF investments only through direct plans
8. Reports to clients at intervals not exceeding 3 months, including commission paid to distributors
9. Auditor's deficiencies must be rectified within 2 months of the report
10. Associate/related-party limits: 15% single associate (equity, or debt/hybrid), 25% across associates, 30% overall (not applicable to MF investments)

11 Regulatory sandbox exemption up to 12 months

12 Code of conduct: PM must obtain in writing the client's interests in bodies that could give UPSI; media advice requires disclosure of positions

13 Eligible Fund Managers (Chapter III) are exempt from HWM, fiduciary, ₹50 lakh minimum etc.; they file quarterly reports to SEBI

14 PMS distributors must register with APMI and never assure returns

15 Soft-dollar arrangements should be avoided; ROS requires analyst pay not linked to investment banking

16 CSCRF: PMs with AUM above ₹3,000 crore are mid-size REs; ≤ ₹3,000 crore are self-certification REs; fewer than 100 clients exempt from M-SOC; cyber incidents reported within 6 hours (email) and 24 hours (portal)

PMLA cash threshold **> ₹10 lakh**	Pre-cleared trade window **≤ 7 trading days**	Contra trade bar **≥ 6 months**	Unlisted (non-disc/advisory) **Up to 25% of AUM**
Associate limits **15% single / 25% multiple / 30% overall**	Client report frequency **Not exceeding 3 months**	Fix audit deficiencies **Within 2 months**	
Sandbox exemption **Up to 12 months**	Cyber incident reporting **6 hours email / 24 hours portal**		

🎯 EXAM TIP
Expect the 15/25/30 related-party table, the 25% unlisted rule, PMLA thresholds and several 'which is a fraudulent practice' items.

⚠️ TRAP
The 25% unlisted limit applies to non-discretionary and advisory services only; discretionary PMs invest in listed or traded securities (except for large value accredited investors).



Chapter 12 · Weight: 5 marks

Introduction to Indices

What indices are for, and the four weighting schemes (price, value/free-float, equal, fundamental), divisor adjustments, index methodologies (Sensex, NIFTY 50), bond indices and hybrid composites.

MAKE IT STICK
A price-weighted index is a class where the tallest kid's mood decides the class photo. A value-weighted index lets the biggest kids matter most. An equal-weighted one gives everyone the same say.

1. Uses: market gauge, benchmark, basis for index funds, proxy for the market portfolio (beta)
2. Dow Jones Average (1884) was the first index and is price-weighted
3. Price-weighted: sum of prices ÷ divisor; high-priced stocks dominate; the divisor is adjusted for splits (downward bias for growth stocks that split)
4. Value-weighted: weights by (free-float) market cap; auto-adjusts for splits; large caps dominate
5. Sensex moved to free float on 1 September 2003; NIFTY 50 on 26 June 2009
6. Free float excludes promoter, government-promoter, FDI, strategic, cross-holdings, ESOP trust and locked-in shares
7. Equal-weighted: arithmetic average of returns; no size bias but high turnover and rebalancing cost
8. Fundamental-weighted: weights by sales, book value, dividends etc.; counters overweighting of overvalued stocks
9. NIFTY 50 eligibility: 100% trading frequency over 6 months; impact cost ≤ 0.50% for a ₹10 crore portfolio in 90% of observations; free-float cap at least 1.5× the smallest constituent
10. Sensex: 30 companies from S&P BSE 100; 6-month listing history
11. NIFTY G-Sec indices: 6 duration buckets; bonds above ₹5,000 crore outstanding; rebalanced monthly
12. NIFTY 50 Hybrid Composite Debt indices blend NIFTY 50 TR with debt (e.g. 70:30, 65:35, 50:50, 15:85); reset monthly

First index	NIFTY 50 impact cost	Sensex free float from	NIFTY 50 free float from
Dow Jones Average, 1884	**≤ 0.50% for ₹10 crore**	**1 September 2003**	**26 June 2009**

🎯 EXAM TIP
Calculations: divisor after a split, equal-weighted and fundamental-weighted index changes.

⚠️ TRAP
After a split in a price-weighted index, the index value stays the same and the divisor changes.



Chapter 13 · Weight: 5 marks

Concept of Informational Efficiency

Operational vs informational efficiency, random walk, Fama's three forms of EMH, anomalies (January, size, value) and what efficiency means for technical analysis, fundamental analysis and index funds.

MAKE IT STICK
In an efficient market, finding a ₹500 note on a crowded street is nearly impossible: someone else picked it up seconds ago. The paradox is that the street stays clean only because so many people keep looking.

1. Operational efficiency = low transaction costs; informational efficiency = prices fully reflect information
2. Efficient price is an unbiased estimate of value; deviations are random
3. Fama (1970) formalised EMH as a fair game model
4. Weak form: past price/volume data reflected, so technical analysis is useless
5. Semi-strong form: all public information reflected, so fundamental analysis on public data cannot earn abnormal returns
6. Strong form: public plus private (insider) information reflected
7. Anomalies: January effect (tax-loss selling), size effect (small firms earn more), value effect (high book-to-price earns more)
8. Internal contradiction: markets become efficient only because participants believe they are not
9. Efficiency supports index funds: be with the market and minimise transaction cost and tax

Weak form kills	Semi-strong kills	Strong form includes
Technical analysis	**Fundamental analysis on public info**	**Insider information**

🎯 EXAM TIP
Map each anomaly or strategy to the EMH form it violates.

⚠️ TRAP
Each stronger form includes the weaker ones; semi-strong already includes weak.



Chapter 14 · Weight: 5 marks

Behavioural Finance

Why real investors depart from rational models: bounded rationality, prospect theory, emotional biases vs cognitive errors, fusion investing and the psychology of bubbles and crashes.

MAKE IT STICK
Holding a losing stock 'until it comes back' while quickly selling a winner to 'lock in profit' is the disposition effect: a ₹1,000 loss hurts more than a ₹1,000 gain pleases.

1. Standard finance assumes rational, risk-averse, self-interested investors with full information
2. Kahneman (2002) and Thaler (2017) Nobel prizes brought behavioural finance forward
3. Bounded rationality (Herbert Simon): people 'satisfice' rather than optimise
4. Prospect theory (Kahneman and Tversky, 1979): reference point; risk-averse in gains, risk-seeking in losses; losses hurt more
5. Emotional biases: loss aversion, stereotyping, overconfidence, endowment, status quo / regret aversion
6. Cognitive errors: mental accounting (Thaler), framing, anchoring

7 Disposition effect (Shefrin and Statman, 1985): sell winners early, hold losers too long

8 Regret from errors of commission is more intense than from omission, which drives status quo bias

9 Fusion investing combines value/growth fundamentals with momentum: value screen, then quality (e.g. Piotroski score 0 to 9), then momentum

10 Bubbles are fuelled by liquidity, celebrity status, momentum (herding) and illusion of control

Prospect theory	Bounded rationality	Disposition effect	Piotroski score range
Kahneman and Tversky, 1979	**Herbert Simon**	**Shefrin and Statman, 1985**	**0 to 9**

🎯 EXAM TIP
Identify the bias from a short story; know which biases are emotional vs cognitive.

⚠️ TRAP
Mental accounting, framing and anchoring are COGNITIVE errors; loss aversion and overconfidence are EMOTIONAL biases in this workbook.



Chapter 15 · Weight: 5 marks

Introduction to Modern Portfolio Theory

Markowitz's mean-variance framework: expected return and variance of a security and a portfolio, covariance and correlation, the efficient frontier, utility and estimation issues.

MAKE IT STICK
Mixing an umbrella seller with an ice-cream seller smooths your income whatever the weather. That's negative correlation reducing portfolio risk without giving up average return.

1 Markowitz 'Portfolio Selection' (Journal of Finance, 1952); Nobel 1990

2 Investors maximise return for a given risk and decide solely on expected return and risk

3 Expected return = Σ probability x return; portfolio return = Σ weight x return

4 Portfolio variance depends on weights, individual risks and covariances (correlations)

5 Two assets: $\sigma p^2 = w1^2\sigma1^2 + w2^2\sigma2^2 + 2w1w2\rho\sigma1\sigma2$

6 Covariance = ρ x σ1 x σ2; correlation ranges from -1 to +1

7 If ρ < 1 then portfolio σ < weighted average σ (diversification); ρ = -1 can give zero risk

8 Efficient frontier: highest return for each risk level (umbrella-shaped curve)

9 Utility $U = E(r) - \frac{1}{2}A\sigma^2$; invest if utility exceeds the risk-free rate

10 Number of correlation estimates = $(n^2 - n)/2$, e.g. 1,225 for 50 securities

11 Inputs: expected returns, standard deviations and the correlation matrix; errors are estimation risk

Two-asset variance	Utility	Correlations needed
$w1^2\sigma1^2 + w2^2\sigma2^2 + 2w1w2\rho\sigma1\sigma2$	$E(r) - 0.5 \times A \times \sigma^2$	$(n^2 - n)/2$

🎯 EXAM TIP
Two-asset portfolio risk, expected return from scenarios, and utility-based accept/reject decisions.

⚠️ TRAP
Portfolio standard deviation is NOT the weighted average of individual SDs unless ρ = +1.

Chapter 16 · Weight: 5 marks

Introduction to Capital Market Theory

Adding a risk-free asset turns the efficient frontier into the Capital Market Line; the market portfolio removes unsystematic risk; CAPM and the Security Market Line price systematic risk; APT extends to multiple factors.

MAKE IT STICK
With a risk-free asset you only need one risky 'super-portfolio' M. A cautious investor mixes M with T-bills, an aggressive one borrows to buy more M. Everyone just turns the risk dial on the same line.

1. CAPM developed by Sharpe (1964), Treynor (1962), Lintner (1965) and Mossin (1966)
2. Assumptions include borrowing/lending at Rf, homogeneous expectations, one-period horizon, no taxes or costs, equilibrium
3. Risk-free asset: zero SD and zero correlation with risky assets
4. CML: $E(Rp) = Rf + (Rm - Rf) \times \sigma p/\sigma m$; portfolios on the CML are perfectly correlated with M
5. Lending portfolios lie left of M, borrowing portfolios right of M
6. Market portfolio M contains all risky assets in market-value proportions, so its correlation with a completely diversified portfolio is +1
7. Diversification benefits largely end after about 20 to 30 securities
8. CAPM: $E(Ri) = Rf + \beta i\,(E(Rm) - Rf)$; $\beta = Cov(i,m)/\sigma m^2$; market beta = 1
9. SML plots return against beta; above SML = undervalued, below = overvalued
10. Beta is unstable for single stocks but stable for portfolios (over 50 stocks, over 26 weeks); betas regress toward 1
11. APT (Stephen Ross): multiple factor betas; does not need a market portfolio or normal returns

CAPM	CML x-axis	SML x-axis	Workbook SML example
$Rf + \beta(Rm - Rf)$	Total risk (σ)	Beta	8% + 1.5 x 14% = 29%

🎯 EXAM TIP
Undervalued/overvalued via SML, CAPM numericals and CML vs SML distinctions.

⚠️ TRAP
CML uses standard deviation and holds only for efficient portfolios; SML uses beta and holds for every security.

Chapter 17 · Weight: 10 marks

Risk

The risk management process, the zoo of risks (market, non-market, liquidity, operational, regulatory, legal, geopolitical, currency, country, concentration, pandemic) and how to measure them: beta, VaR (parametric, historical, Monte Carlo), stress tests, bid-ask liquidity cost and credit risk (PD, EAD, LGD).

MAKE IT STICK
VaR is a weather forecast for losses: '95% of days, you won't lose more than ₹1.2 million.' It says nothing about how bad the other 5% storms can be, which is why stress tests exist.

1. Process: set objectives, identify, analyse, evaluate (severity and frequency), treat (tolerate, mitigate, transfer, terminate), control and monitor
2. Market (systematic) risk is hedged with derivatives; non-market risk is diversified away
3. Company-specific risk = business risk + financial risk
4. Diversification benefit flattens after about 30 to 35 stocks
5. Regulatory risk is a part of legal risk; legal risk relates to enforceability of contracts
6. VaR: maximum loss over a period, at a confidence level, in normal market conditions
7. Parametric VaR = [E(daily return) x n - z x σdaily x √n] x portfolio value; 250 or 252 trading days
8. One-tailed z: 1.282 (90%), 1.645 (95%), 2.33 (99%); 1.96 is for a two-sided 95% range
9. Historical simulation VaR is usually lower than parametric; Monte Carlo (~10,000 runs) handles complex portfolios and any distribution
10. VaR limitations: subjectivity; underestimates extreme losses and gains
11. Sensitivity measures: beta (equity), duration (bonds), delta/gamma/vega (options); stress test = extreme scenario
12. Liquidity cost = (ask - bid)/mid x mid value; liquidity-adjusted VaR = VaR + cost of unwinding
13. Credit risk: probability of default, exposure at default, loss given default; four Cs: capacity, collateral, covenants, capital

z 95% one-tail	z 99% one-tail	z 90% one-tail	Daily σ	Monte Carlo runs
1.645	2.33	1.282	Annual σ ÷ √250	~10,000

Risk treatments
Tolerate, Mitigate, Transfer, Terminate

🎯 **EXAM TIP**
The VaR caselet (daily to n-day scaling) is a workbook favourite; also interpreting a VaR statement.

⚠️ **TRAP**
Scale return by n but standard deviation by √n. Use 1.645 (not 1.96) for 95% VaR.



Chapter 18 · Weight: 15 marks

Equity Portfolio Management Strategies

Passive (buy and hold, indexing by full replication or sampling) vs active (market timing, sector rotation), the fundamental law of active management, smart beta and factor models, momentum, growth/value/blend styles, ESG screening, core-satellite, alpha-beta separation, derivatives and global investing.

MAKE IT STICK
Core-satellite is a thali: the rice and dal (70 to 80% passive core) keep you fed cheaply, while the small spicy pickles (active satellites) add the flavour, and the cost.

1. Passive aims to match the benchmark; active aims to beat it on a risk-adjusted basis at higher cost
2. Full replication suits small, liquid indices (NIFTY 50, Sensex); sampling suits large indices (NIFTY 500)
3. Market timers raise beta or shift to stocks when expecting a rise; sector rotation is a top-down strategy
4. Fundamental law (Grinold and Kahn, 1989): IR = IC x √Breadth
5. IC = 2 x (right predictions ÷ total) - 1; ranges from -1 to +1; IC = 0 is a coin toss
6. Breadth = independent forecasts per period x number of periods; doubling breadth raises IR by √2 (1.414×)
7. Smart beta: rule-based like passive, factor choice like active (quality, value, low volatility, momentum)
8. NSE multi-factor indices (July 2017): Alpha Low-Vol 30, Quality Low-Vol 30, Alpha Quality Low-Vol 30, Alpha Quality Value Low-Vol 30
9. Factor models: macroeconomic, fundamental, statistical
10. Growth: high earnings growth, low payout, high valuation; value: low P/E, P/B, P/S, PEG, high dividend yield

11 SRI screens: negative (exclude sin stocks), positive (rank on ESG), best-in-class (best per sector)

12 Core-satellite: about 70 to 80% core (passive, minimise tracking error), rest satellite (active)

13 Alpha-beta separation / portable alpha: beta from passive exposure, alpha from market-neutral long-short

14 Protective put caps downside at the premium; selling index futures hedges a diversified portfolio

15 Market cap to GDP: below 50% undervalued; 50 to 75% moderately undervalued; 75 to 90% fair; 90 to 110% moderately overvalued; above 110% overvalued

16 Home bias reasons: familiarity, optimism, risks, predictability; costs: concentration and missed diversification

Fundamental law	IC	Core share	Mcap/GDP fair value band
IR = IC x √Breadth	**2 x (right/total) – 1**	**70% to 80%**	**75% to 90%**

US share of world mcap
> 50%

🎯 EXAM TIP
IC/IR caselets, protective put pay-offs, style identification and 'which is NOT passive' questions.

⚠️ TRAP
Smart beta, factor, growth, value and momentum portfolios are NOT passive strategies; buy and hold and indexing are.



Chapter 19 · Weight: 15 marks

Fixed Income Portfolio Management Strategies

Passive bond strategies (buy and hold, indexing, immunization), active strategies (duration calls, barbell/bullet, floaters, maturity extension, roll-down, convexity, credit and spread analysis), global bonds and derivatives (IRS, swaptions, futures, forwards, CDS, currency swaps).

MAKE IT STICK
Immunization is balancing a see-saw: if rates fall, bond prices rise but reinvested coupons earn less; if rates rise, the reverse. Set the duration equal to your horizon and the two sides cancel out.

1 Passive strategies: buy and hold, indexing, immunization

2 Buy and hold locks yield and removes price risk but not reinvestment risk

3 Tracking error causes: costs, similar-not-same securities, weight differences, illiquid securities, concentration limits, cash holdings

4 Benchmark choice: risk-free long-term = G-Sec index; short-term cash = liquid index; income = long duration + aggregate; low risk = low duration + high credit quality; inflation = floating-rate benchmark

5 Immunization: Macaulay duration of the portfolio = liability horizon; zero-coupon bond with matching maturity is the ideal

6 Immunization needs periodic re-balancing as rates change, so it is not as passive as buy and hold

7 Expect rates up: cut duration; expect rates down: raise duration

8 Barbell (e.g. 40% short / 20% mid / 40% long) when the rate direction is uncertain; durations above 7 years long, 3 to 7 mid, below 3 short

9 Floater's duration is only the time to the next reset

10 Roll-down: on an upward-sloping curve, yield falls as maturity shortens, so price rises

11 Positive convexity: price rises more and falls less than duration predicts

12 Yield curve 3 S's: shift, slope, shape

13 Credit spreads widen in recessions and narrow in expansions; term spread = long yield minus short yield

14 Credit research uses the four Cs to estimate PD and LGD; credit ratings are mandatory before issuing bonds or CP in India

15 CDS = insurance against a credit event; swaption = option to enter an IRS; cross-currency swap exchanges floating payments in two currencies

Long / mid / short duration	Barbell example	Zero-coupon PV example
> 7 / 3 to 7 / < 3 years	**40 / 20 / 40**	**₹10 lakh in 10 yrs at 10% = ₹3,85,543**

Immunization rule
Duration = liability horizon

🎯 EXAM TIP
Strategy selection for a rate view, immunization weights from two durations, convexity statements and benchmark matching.

⚠️ TRAP
Credit risk is NOT a significant risk for G-Secs, but price, reinvestment and liquidity risk still are.

Chapter 20 · Weight: 10 marks

Performance Measurement and Evaluation of Portfolio Managers

Every return measure (HPR, TWRR, MWRR/IRR, AM vs GM, gross vs net, pre vs post tax, CAGR, cash drag, alpha), risk measures (SD, semi-variance, beta, tracking error), risk-adjusted ratios (Sharpe, Treynor, Sortino, Information Ratio, M^2), benchmarking, attribution analysis, SEBI reporting rules and GIPS.

MAKE IT STICK

TWRR grades the driver; MWRR grades the trip. If the passenger keeps adding fuel just before uphill stretches, the trip looks worse, but that's not the driver's fault. So SEBI judges managers on TWRR.

1. HPR = (Income + End - Begin) ÷ Begin
2. TWRR chain-links sub-period returns, removes the effect of client cash flows; mandated by GIPS and SEBI for PM performance
3. Sub-period return = (EMV - BMV - contribution + withdrawal) ÷ (BMV + contribution - withdrawal)
4. MWRR = IRR, depends on timing of cash flows
5. Investors get XIRR with min, max and median XIRR across investors; IA performance uses TWRR with the benchmark's trailing return
6. Geometric mean ≤ arithmetic mean (equal only if all returns are equal)
7. Post-tax = pre-tax x (1 - t); pre-tax = post-tax ÷ (1 - t)
8. Cash drag: include cash and liquid funds in return calculations
9. Performance must be net of all fees and expenses (including taxes); marketing must carry a 'not verified by SEBI' disclaimer
10. Jensen alpha = Rp - [Rf + β(Rm - Rf)]
11. Beta = Cov(p,m)/Var(m) = ρ x σp/σm; portfolio beta is the weighted average
12. Sharpe = (Rp - Rf)/σp for poorly diversified investors; Treynor = (Rp - Rf)/β for well-diversified investors
13. Sortino uses downside (semi) deviation; Information Ratio = active return ÷ tracking error
14. M^2 = (Rp - Rf) x σm/σp + Rf, compared with Rm
15. Attribution: allocation Σ Rpi(Wai - Wpi), selection Σ Wpi(Rai - Rpi), interaction Σ(Wai - Wpi)(Rai - Rpi)
16. APMI prescribes valuation norms and empanels agencies for debt/money market valuation; PaRRVA verifies past risk-return claims
17. GIPS advertising: 1, 3, 5 year annualised returns; periods under 1 year must not be annualised

Sharpe	Treynor	Information ratio	M^2
(Rp – Rf)/σp	**(Rp – Rf)/βp**	**(Rp – Rb)/tracking error**	**(Rp – Rf) x σm/σp + Rf**

Workbook TWRR example	SEBI performance measure
0.98 x 1.08 x 1.07 – 1 = 14.71%	**TWRR (IA) and XIRR (investor)**

🎯 EXAM TIP
This chapter supplies most caselets: TWRR vs MWRR, Sharpe vs Treynor choice, M², Jensen alpha, attribution decomposition, gross vs net return.

⚠️ TRAP
Choose Treynor only when the investor's overall wealth is well diversified; otherwise Sharpe. Arithmetic average over-states multi-year performance.

Chapter 21 · Weight: 5 marks

Portfolio Rebalancing

Why portfolios drift, the cost-benefit of rebalancing, calendar vs threshold rules, and three dynamic strategies: buy and hold (do nothing), constant mix (do something) and CPPI (floor plus multiplier).

MAKE IT STICK
CPPI is a trapeze act with a safety net. The net is the floor; the height you climb is the cushion times your courage (multiplier). Fall towards the net and you automatically climb down.

1. Rebalancing is needed due to market drift and changes in investor circumstances; the IPS should hold a rebalancing policy
2. Costs: monitoring, transaction costs and taxes on selling appreciated assets; illiquid assets are harder to rebalance
3. Calendar (time-based) rebalancing is simple; quarterly is popular but ignores how far the mix has drifted
4. Threshold (corridor) rebalancing: e.g. 60% equity target with 50% and 70% triggers; tighter control but needs monitoring
5. Higher transaction cost assets get wider corridors; more volatile assets get narrower ones
6. Buy and hold: do nothing; floor from safe assets; unlimited upside; market risk tends to increase
7. Constant mix: periodically restore weights (sell winners, buy losers); maintains market risk exposure
8. CPPI: risky exposure = m x (portfolio value - floor); floor = PV of protected amount at Rf; multipliers usually 3 to 5

CPPI exposure	Typical multiplier	Threshold example
m x (Portfolio – Floor)	3 to 5	60% target, 50% / 70% triggers

Workbook CPPI
₹150 cr portfolio, ₹100 cr floor, m = 3 → ₹150 cr risky

🎯 EXAM TIP
CPPI allocations (including floor = PV of capital), constant-mix buy/sell direction and which strategy has unlimited upside.

⚠️ TRAP
In CPPI the floor is the PRESENT VALUE of the protected amount, not the amount itself.

🧮 Formulas

Nominal required rate

```
(1 + real rate) x (1 + inflation) - 1
```

Real 3%, inflation 5%: 1.03 x 1.05 - 1 = 8.15%

Current yield

```
Annual coupon ÷ market price
```

₹9 coupon, price ₹112.50: 8%

Modified duration

```
Macaulay duration ÷ (1 + YTM/frequency)
```

MacD 6, YTM 10% semi-annual: 6/1.05 = 5.71

Bond price change

```
%ΔP ≈ -MD x Δy
```

MD 5.71, yield +0.50%: -2.86%

Dividend discount model

```
P = D1 / (k - g), D1 = D0(1 + g)
```

D0 ₹4, g 5%, k 12%: P = 4.20/0.07 = ₹60

PEG ratio

```
(P/E) ÷ EPS growth rate (%)
```

P/E 30, growth 20%: PEG 1.5, looks overvalued

Enterprise value

```
Market cap + market value of debt -
excess cash
```

800 + 300 - 100 = ₹1,000 crore

WACC

```
Ke x We + Kd x (1 - t) x Wd
```

14% x 0.6 + 10% x 0.75 x 0.4 = 11.4%

Expected return (scenarios)

```
Σ pᵢ x Rᵢ
```

0.2 x 30% + 0.5 x 12% + 0.3 x (-5%) = 10.5%

Two-asset portfolio risk

```
σp² = w1²σ1² + w2²σ2² + 2w1w2ρσ1σ2
```

60/40, σ 20%/10%, ρ 0.3: σp = 13.74%

Utility

```
U = E(r) - 0.5 x A x σ²
```

E(r) 14%, σ 20%, A = 4: U = 14% - 8% = 6%

CAPM / SML

```
E(Ri) = Rf + βi (Rm - Rf)
```

Rf 7%, β 1.4, MRP 8%: 18.2%

CML

```
E(Rp) = Rf + (Rm - Rf) x σp/σm
```

Rf 6%, Rm 14%, σm 16%, σp 8%: 10%

Beta

```
Cov(p,m)/σm² = ρ x σp/σm
```

ρ 0.8, σp 27%, σm 18%: β = 1.2

Parametric VaR

```
[μdaily x n - z x σdaily x √n] x Portfolio value
```

μ 20% p.a., σ 25%, 95%, 1 day: 0.08% - 1.645 x 1.58% = -2.52%

Liquidity cost

```
(Ask - Bid)/Mid x mid-market value
```

Bid 198, ask 202, 50,000 shares: 2% x ₹1 crore = ₹2 lakh

Information coefficient

```
IC = 2 x (right ÷ total) - 1
```

60 right of 80: IC = 0.5

Fundamental law

```
IR = IC x √Breadth
```

IC 0.5, breadth 100: IR = 5

Market cap to GDP

```
Total listed market cap ÷ GDP x 100
```

380 ÷ 400 = 95%: moderately overvalued

Holding period return

```
(I + E - B) ÷ B
```

(1 + 57 - 50) ÷ 50 = 16%

TWRR

```
Π(1 + sub-period return) - 1
```

(1.10 x 1.0593) - 1 = 16.52%

Sub-period return

```
(EMV - BMV - C + W) ÷ (BMV + C - W)
```

(57.2 - 44 - 10) ÷ (44 + 10) = 5.93%

Geometric mean / annualised

```
[Π(1 + r)]^(1/n) - 1
```

12%, -8%, 20%: 7.33% p.a.

CAGR

```
(End/Begin)^(1/t) - 1
```

₹50 lakh to ₹80 lakh in 5 years: 9.86%

Post-tax return

```
Pre-tax x (1 - t)
```

12% at 20% tax: 9.6%

Jensen alpha

```
Rp - [Rf + β(Rm - Rf)]
```

18% - [6% + 1.2 x 8%] = 2.4%

Sharpe ratio

```
(Rp - Rf)/σp
```

(16 - 6)/20 = 0.5

Treynor ratio

```
(Rp - Rf)/βp
```

(16 - 6)/1.25 = 8

Sortino ratio

```
(Rp - Rf)/downside deviation
```

(16 - 6)/8 = 1.25

Information ratio

```
(Rp - Rb)/σ(Rp - Rb)
```

Active return 3%, tracking error 2%: 1.5

M² measure

```
(Rp - Rf) x σm/σp + Rf
```

(20 - 6) x 18/25 + 6 = 16.08% vs market 15%

Attribution: allocation

```
Σ Rpi x (Wai - Wpi)
```

10 x 0.1 + 6 x (-0.1) = 0.4%

Attribution: selection

```
Σ Wpi x (Rai - Rpi)
```

0.6 x 2 + 0.4 x (-1) = 0.8%

Attribution: interaction

```
Σ (Wai - Wpi)(Rai - Rpi)
```

0.1 x 2 + (-0.1)(-1) = 0.3%

Foreign currency return

```
(1 + local return) x (FX end/FX start)
- 1
```

10% in USD, ₹80 to ₹84: 15.5%

Cash-drag adjusted return

```
Σ (weight x return) on total capital
```

80% at 15% + 20% at 5% = 13%

CPPI risky exposure

```
m x (Portfolio value - Floor)
```

₹200 lakh, floor ₹160 lakh, m = 3: ₹120 lakh

Zero-coupon immunization price

```
FV ÷ (1 + r)^n
```

₹50 lakh, 8 yrs at 8%: ₹27.01 lakh

Index divisor after split

```
New divisor = new sum of prices ÷ old
index value
```

Sum 160, index 50: divisor 3.2

High water mark fee

```
Fee % x (pre-fee value - highest
earlier value)
```

20% x (1.45 - 1.30) crore = ₹3 lakh

🔢 Numbers to remember

90 MCQ (1 mark) + 6 caselets x 5 sub-q (2 marks) = 150 marks Exam	**180 minutes / 60% (90 marks)** Duration / pass	**25% of the question's marks** Negative marking
₹5 crore PM net worth	**₹50 lakh (cash and/or securities)** Minimum client investment	**5 years (incl. 2 years in PM/IA/fund management)** Principal Officer experience
Graduate + 2 years' experience Additional employee	**3% yr 1, 2% yr 2, 1% yr 3, nil after** Exit load caps	**0.50% p.a. of average daily AUM (excl. brokerage)** Operating expenses cap
Up to 25% of AUM Unlisted (non-disc/advisory)	**Up to 100%** Large value accredited investors, unlisted	**15% single, 25% across, 30% overall** Associate investment limits
At least every 3 months Client reports	**Within 2 months** Audit deficiencies fixed	**Within 6 months of year end** Audit certificate to SEBI
5 accounting years Books preserved	**AUM ≥ ₹1,000 crore** Automated dealing system	**Within 7 working days** STR to FIU-IND
> ₹10 lakh (or monthly series > ₹10 lakh) PMLA cash reporting	**≤ 7 trading days** Pre-clearance validity	**≥ 6 months** Contra-trade restriction
Up to 12 months Regulatory sandbox	**6 hours (email), 24 hours (portal)** Cyber incident reporting	**AUM ≤ ₹3,000 crore; < 100 clients exempt from M-SOC** CSCRF self-certification RE
182 days, or 60 days + 365 in previous 4 years Resident (individual)	**Indian income > ₹15 lakh, not taxed elsewhere** Deemed resident	**> 12 months / > 24 months** LTCG listed / unlisted
≥ 25 members, ≤ 10% each, ≤ 5% residents, ≤ 20% per entity, ≥ ₹100 crore corpus Section 9A	**USD 2,50,000 per financial year** LRS	**90%: 1.282, 95%: 1.645, 99%: 2.33** z-scores (one tail)
250 or 252 Trading days	**~20 to 30 stocks (Ch 16), 30 to 35 (Ch 17)** Diversification limit	**70% to 80%** Core in core-satellite

Usually 3 to 5
CPPI multiplier

Short < 3, mid 3 to 7, long > 7 years
Duration buckets

75% to 90%
Mcap/GDP fair band

Max 3 per strategy
APMI benchmarks

~40% across funds; ~90% within a fund over time
Asset allocation explains

📖 Glossary

Discretionary PMS	Manager exercises any degree of discretion over investments for each client individually.
Non-discretionary PMS	Client decides what and when; the manager only executes.
Principal Officer	Employee responsible for investment decisions and overall supervision of the PM's operations.
Investment Approach (IA)	Documented broad outlay of securities and instruments the PM will invest in for a client.
Disclosure Document	Mandatory document (static and dynamic parts) given to prospective clients before or at signing.
MITC	Most Important Terms and Conditions document every PMS client must receive.
High Water Mark	Highest portfolio value reached; performance fee is charged only above it.
Hurdle rate	Minimum return that must be beaten before a performance fee applies.
Accredited investor	Investor to whom the ₹50 lakh minimum need not apply, subject to disclosures and agreement.
Co-investment PM	PM exempted from several rules (minimum ticket, net worth, separate compliance officer, custodian).
Eligible Fund Manager	India-based manager of offshore 'eligible investment funds' under Section 9A and Chapter III of the PM Regulations.
SMART ODR	Online dispute resolution portal linking investors to MII-administered conciliation and arbitration.
CKYCR	Central KYC Records Registry run under CERSAI's template.
IPS	Investment Policy Statement: the client's objectives, constraints and evaluation framework.
SAA / TAA	Strategic (long-term policy) vs tactical (short-term opportunistic) asset allocation.
Free float	Shares available for public trading, excluding promoter, strategic and locked-in holdings.
Efficient frontier	Set of portfolios with maximum return for each level of risk.
Market portfolio (M)	All risky assets in market-value proportions; only systematic risk remains.
Beta	Sensitivity of a security's return to the market; measure of systematic risk.
Tracking error	Standard deviation of the portfolio's returns minus benchmark returns (active risk).
Information coefficient	Correlation between forecast and actual outcomes; a manager's skill.

Smart beta	Rule-based, factor-weighted index strategies between passive and active.
Portable alpha	Separating alpha (market-neutral bets) from beta exposure, often across asset classes.
Immunization	Matching portfolio duration to liability horizon so price and reinvestment effects offset.
Convexity	Second-order price effect; positive convexity means bigger gains and smaller losses than duration predicts.
Swaption	Option to enter into an interest rate swap.
Credit Default Swap	Protection against a credit event on a bond in return for a premium.
VaR	Estimated maximum loss over a period at a confidence level in normal markets.
TWRR	Time-weighted return; removes client cash-flow effects; measures manager skill.
MWRR / XIRR	Money-weighted (IRR) return; reflects the investor's actual timing of cash flows.
M^2 (Modigliani)	Portfolio return re-scaled to the market's risk, compared directly with the market return.
CPPI	Constant Proportion Portfolio Insurance: risky exposure = multiplier x (value - floor).
Soft dollars	Paying higher brokerage in exchange for research or other services; should be avoided.
PaRRVA	Past Risk and Return Verification Agency, SEBI's tech-based check on past-return claims.

⚡ Quick revision: the night before

1. Exam: 90 MCQ x 1 + 30 caselet sub-questions x 2 = 150 marks, 3 hours, pass 90 marks, minus 25% per wrong answer.
2. PM must be a body corporate with ₹5 crore net worth; minimum client ticket ₹50 lakh.
3. Principal Officer: qualification + 5 years' experience (2 in PM/IA/fund mgmt) + NISM certificate.
4. Advisory-only PMs need no custodian; everyone else appoints one.
5. Exit load: max 3%, 2%, 1% in years 1, 2, 3; nil after; no lock-in; no upfront fee.
6. Operating expenses (excluding brokerage) ≤ 0.50% p.a. of average daily AUM; brokerage at actuals.
7. Performance fee only above the high water mark (and hurdle).
8. Discretionary: listed/traded securities, money market, MF (direct plans). Non-discretionary/advisory: up to 25% unlisted.
9. Associates: 15% single, 25% multiple, 30% total of client AUM; not for MF investments.
10. Reports to clients at least quarterly; fix auditor's deficiencies within 2 months.
11. STR to FIU-IND within 7 working days; PMLA cash reporting above ₹10 lakh.
12. Pre-cleared trades within 7 trading days; no contra trade for 6 months.
13. AUM ≥ ₹1,000 crore needs an automated order and allocation system.
14. IPS is the first step; SAA = time in the market; TAA = timing the market.
15. BB&K: Adventurer, Celebrity, Individualist, Guardian, Straight Arrow.
16. Life cycle: accumulation, consolidation, spending, gifting.
17. Resident: 182 days, or 60 days + 365 days in 4 prior years; deemed resident = NOR.
18. Listed shares long term after 12 months; unlisted after 24 months.
19. Section 9A: 25+ members, ≤10% per member, ≤5% resident, ≤20% per investee, ≥₹100 crore corpus, manager ≤20% profits.
20. Weak EMH kills technical analysis; semi-strong kills fundamental analysis on public info.
21. Mental accounting, framing, anchoring = cognitive; loss aversion, overconfidence, endowment, status quo = emotional.
22. $\sigma p <$ weighted average σ whenever $\rho < +1$.
23. CML uses σ (efficient portfolios); SML uses β (all securities). Above SML = undervalued.
24. VaR z: 1.282 / 1.645 / 2.33; scale σ by $\sqrt{n}$ and return by n.
25. IR = IC x √Breadth; IC = 2p - 1.
26. Core-satellite: 70 to 80% passive core.
27. Mcap/GDP: <50% undervalued, >110% overvalued.
28. Rates up → shorten duration; uncertain → barbell; immunize: duration = horizon.
29. Positive convexity: price rises more, falls less than duration predicts.
30. TWRR for manager skill and SEBI/GIPS; XIRR for the individual investor.
31. Sharpe for undiversified investors, Treynor for well-diversified ones.
32. $M^2 = (Rp - Rf) \times \sigma m/\sigma p + Rf$; compare with Rm.
33. Attribution: allocation (Rpi x ΔW), selection (Wpi x ΔR), interaction (ΔW x ΔR).
34. Performance reported net of all fees and expenses; disclaimer 'not verified by SEBI'.
35. CPPI risky = m x (value - floor); floor = PV of protected capital; m usually 3 to 5.
36. Constant mix sells winners and buys losers; buy and hold has unlimited upside.

🗓️ 21-day study plan (2 h/day)

Days 1–3: Foundations (Ch 1 to 6)

- Skim Ch 1 to 6; you already know MF basics from V-A
- Practise nominal rate, DDM, PEG, current yield and modified duration
- Take Mock 1

Days 4–8: PMS regulation (Ch 7, 8, 11)

- Make a one-page table of every number: ₹5 crore, ₹50 lakh, 3/2/1 exit load, 0.50%, 25%, 15/25/30
- Work the performance-fee and high water mark examples in Box 8.1 in Excel
- Read the do's and don'ts twice
- Take Mock 2

Days 9–11: Process, tax, indices, EMH, behaviour (Ch 9, 10, 12 to 14)

- Learn BB&K and life-cycle phases with examples
- Memorise residency days and Section 9A numbers
- Practise index divisor and fundamental-weighted index calculations
- Take Mock 3

Days 12–15: Theory and risk (Ch 15 to 17)

- Build a two-asset portfolio risk sheet in Excel
- CAPM, SML over/undervaluation drills
- VaR: 1-day and 10-day at 90/95/99%
- Take Mock 4

Days 16–18: Strategies (Ch 18, 19, 21): 35 marks

- IC, breadth and IR numericals
- Immunization weights, barbell, convexity statements
- CPPI and constant mix drills
- Take Mock 5

Days 19–21: Performance (Ch 20) and full simulation

- TWRR, XIRR, Sharpe, Treynor, M^2, Jensen and attribution in Excel
- Take tough Mocks 6 and 7 under time
- Run the 120-question simulator; revise wrong answers and the quick-revision list

Independent study notes prepared from the official NISM workbook. Verify against the latest workbook on nism.ac.in.